Lab 2: Unity Catalog Objects and Permissions

Description

You will create your own catalog in Unity Catalog using SQL, then add a schema, two tables and a volume. As the owner, you can do everything with these objects without being granted anything.

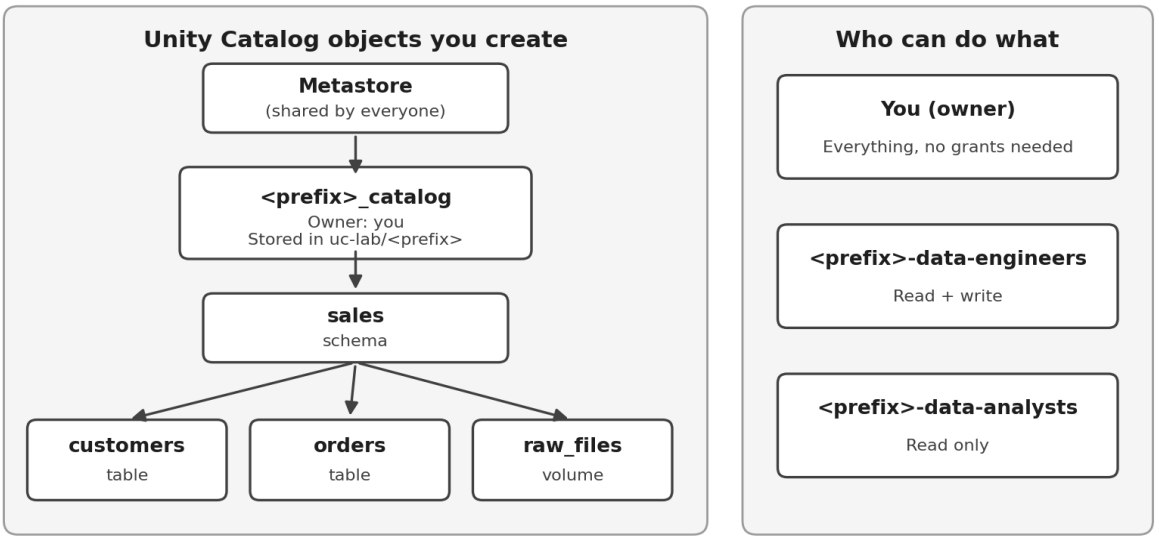
You will then grant read and write access to your Data Engineers group, read-only access to your Data Analysts group, and sign in as each test user from Lab 1 to confirm the access.

Learning Objectives

- Create a catalog, schema, tables and a volume with SQL
- Understand what the owner of an object can do
- Grant Unity Catalog privileges to groups with GRANT and REVOKE
- Verify read + write access and read-only access as non-admin users

Step by Step Guide

What you will build



Your lab values

Use the same prefix as in Lab 1 (your first name in lowercase). In every SQL script below, replace <prefix> with your prefix before running it.

Item	Value
Catalog	<prefix>_catalog (for example bob_catalog)
Schema	sales
Tables	customers, orders
Volume	raw_files
Groups (from Lab 1)	<prefix>-data-engineers, <prefix>-data-analysts
Test users (from Lab 1)	<prefix>-engineer1, <prefix>-analyst1
SQL warehouse	Shared serverless warehouse named by your trainer
Storage path	abfss://uc-lab@<storage-account>.dfs.core.windows.net (storage account name from your trainer)

Tip: Paste a script into the SQL editor, then use Find and Replace to change every <prefix> at once.

Step 1: Open the SQL editor

1. Sign in to the workspace with your learner account.
2. Click SQL Editor in the sidebar and open a new query tab.
3. In the compute drop-down at the top, select the shared SQL warehouse.
4. Run:

```
SELECT current_user();
```

> Check: The result shows your learner account email.

Step 2: Create your catalog and schema

The MANAGED LOCATION tells Unity Catalog where to store the data for your catalog. Each learner uses their own folder, named with their prefix.

```
CREATE CATALOG IF NOT EXISTS <prefix>_catalog
  MANAGED LOCATION
  'abfss://uc-lab@<storage-account>.dfs.core.windows.net/<prefix>'
  COMMENT 'Lab 2 catalog';

CREATE SCHEMA IF NOT EXISTS <prefix>_catalog.sales
  COMMENT 'Sales data for Lab 2';
```

> Check: Open Catalog in the sidebar and refresh. Your catalog appears with three schemas: default, information_schema and sales.

Step 3: Create a volume and upload the CSV file

```
CREATE VOLUME IF NOT EXISTS <prefix>_catalog.sales.raw_files;
```

1. Save the text below as a file named customers.csv on your computer (or use the file shared by your trainer).

```
customer_id,name,city,country
1,Aarav Shah,Mumbai,India
2,Mei Tan,Singapore,Singapore
3,Omar Haddad,Dubai,UAE
4,Grace Wanjiru,Nairobi,Kenya
5,Liam Walker,Sydney,Australia
```

2. In Catalog, open <prefix>_catalog > sales > raw_files.
3. Click Upload to this volume, choose customers.csv and upload.
4. Back in the python editor, run:

```
!ls '/Volumes/s_catalog/sales/raw_files/'
```

> Check: customers.csv is listed.

Step 4: Create two tables

```
-- Table 1: load customers from the CSV in the volume
CREATE TABLE IF NOT EXISTS <prefix>_catalog.sales.customers AS
SELECT * FROM read_files(
  '/Volumes/<prefix>_catalog/sales/raw_files/customers.csv',
  format => 'csv', header => true);

-- Table 2: create orders and insert rows
CREATE TABLE IF NOT EXISTS <prefix>_catalog.sales.orders (
  order_id INT, customer_id INT, amount DECIMAL(10,2), order_date DATE);

INSERT INTO <prefix>_catalog.sales.orders VALUES
  (101, 1, 250.00, '2026-09-01'),
  (102, 2, 120.50, '2026-09-03'),
  (103, 3, 560.00, '2026-09-05'),
  (104, 1,  75.25, '2026-09-10'),
  (105, 5, 310.00, '2026-09-12');
```

> Check: Both tables appear under sales in Catalog. customers has 5 rows and orders has 5 rows.

Step 5: Show that the owner can do everything

Nobody has granted you any privileges on this catalog, yet every statement below works because you created it and are its owner.

```
DESCRIBE CATALOG EXTENDED <prefix>_catalog;

SELECT c.name, c.city, o.amount
FROM <prefix>_catalog.sales.orders o
JOIN <prefix>_catalog.sales.customers c ON o.customer_id = c.customer_id;

UPDATE <prefix>_catalog.sales.orders SET amount = amount + 10 WHERE order_id = 101;
DELETE FROM <prefix>_catalog.sales.orders WHERE order_id = 105;
ALTER TABLE <prefix>_catalog.sales.customers ADD COLUMN segment STRING;
```

> Check: DESCRIBE shows you as the **Owner** and shows your storage folder as the storage root. Every statement succeeds, and orders now has 4 rows.

Step 6: Let your groups use the SQL warehouse

Test users need permission to use compute before they can run any query.

1. Click **SQL Warehouses** in the sidebar and open the shared warehouse.
2. Click **Permissions**.
3. Add <prefix>-data-engineers and <prefix>-data-analysts with **Can use**. Save.

> Check: Both groups are listed with Can use.

Step 7: Check that a new user has no data access yet

1. Open a private browser window and sign in to the workspace as:
   <prefix>-analyst1@saurabhuptut.onmicrosoft.com
2. Open SQL Editor, select the shared warehouse and run:

```
SELECT * FROM <prefix>_catalog.sales.customers;
```

> Check: The query fails with a permission or not-found error. In Unity Catalog, users have no access to data by default. Keep this window open.

Step 8: Grant permissions to your groups

Back in your own window (as the owner), run:

```
-- Data Engineers: read + write
GRANT USE CATALOG ON CATALOG <prefix>_catalog TO `<prefix>-data-engineers`;
GRANT USE SCHEMA, SELECT, MODIFY, CREATE TABLE
  ON SCHEMA <prefix>_catalog.sales TO `<prefix>-data-engineers`;
GRANT READ VOLUME, WRITE VOLUME
  ON VOLUME <prefix>_catalog.sales.raw_files TO `<prefix>-data-engineers`;

-- Data Analysts: read only
GRANT USE CATALOG ON CATALOG <prefix>_catalog TO `<prefix>-data-analysts`;
GRANT USE SCHEMA, SELECT
  ON SCHEMA <prefix>_catalog.sales TO `<prefix>-data-analysts`;
GRANT READ VOLUME
  ON VOLUME <prefix>_catalog.sales.raw_files TO `<prefix>-data-analysts`;

SHOW GRANTS ON SCHEMA <prefix>_catalog.sales;
```

Note: Group names contain hyphens, so they must be wrapped in backticks. Privileges granted on the schema apply to every table inside it, including tables created later.

Privilege	data-engineers	data-analysts
USE CATALOG	Yes	Yes
USE SCHEMA	Yes	Yes
SELECT (read tables)	Yes	Yes
MODIFY (insert, update, delete)	Yes	No
CREATE TABLE	Yes	No
READ VOLUME	Yes	Yes
WRITE VOLUME	Yes	No

> Check: SHOW GRANTS lists both groups with the privileges in the table.

Step 9: Verify as the Data Engineer

1. Open a new private window and sign in as <prefix>-engineer1@saurabhuptut.onmicrosoft.com.
2. Open SQL Editor, select the shared warehouse and run each statement one at a time:

```
-- Should work
SELECT * FROM <prefix>_catalog.sales.customers;
INSERT INTO <prefix>_catalog.sales.orders VALUES (201, 2, 99.00, '2026-09-24');
CREATE TABLE <prefix>_catalog.sales.eng_scratch (id INT);
LIST '/Volumes/<prefix>_catalog/sales/raw_files/';

-- Should fail
DROP TABLE <prefix>_catalog.sales.customers;
```

3. In Catalog, open the raw_files volume and upload any small file, such as a copy of customers.csv renamed to engineer_upload.csv.

> Check: The engineer can read, write, create tables and upload files. DROP TABLE fails because only the owner can drop customers. The engineer owns eng_scratch, so they could drop that one.

Step 10: Verify as the Data Analyst

Go back to the <prefix>-analyst1 window from Step 7 and run each statement one at a time:

```
-- Should work
SELECT * FROM <prefix>_catalog.sales.customers;
SELECT * FROM read_files(
  '/Volumes/<prefix>_catalog/sales/raw_files/customers.csv',
  format => 'csv', header => true);

-- Should fail
INSERT INTO <prefix>_catalog.sales.orders VALUES (301, 3, 10.00, '2026-09-24');
CREATE TABLE <prefix>_catalog.sales.analyst_table (id INT);
```

> Check: The analyst can read tables and files in the volume, but cannot change data or create tables. The same query from Step 7 now works, with no sign-out needed.

Step 11 (Optional): Revoke and re-test

```
REVOKE SELECT ON SCHEMA <prefix>_catalog.sales FROM `<prefix>-data-analysts`;
```

1. As <prefix>-analyst1, run the SELECT on customers again. It fails.
2. As the owner, grant SELECT back to the analysts group.

> Check: Access changes as soon as the grant changes.

Outcomes

- You own a catalog with a schema, two tables and a volume, all created with SQL
- You have seen that the owner can do everything without explicit grants

- Data Engineers can read and write; Data Analysts can only read
- You have confirmed that new users have no data access until privileges are granted